VACUUM AND EVAP LEAK GUIDE

How a leak raises idle on this engine

The 4.0L measures load with a MAP sensor. It does not have a mass-airflow sensor. Air that enters through a cracked elbow, a gasket, or a purge valve still changes manifold pressure. The PCM sees that pressure and adds fuel. The engine still speeds up, because the extra air is real and the throttle plate is supposed to be closed. **A vacuum leak is a strong explanation for a high or unstable idle.** LIKELY mechanism. It is not, by itself, proof that the mixture is lean or that the manifold will glow.

High idle — likely

Extra air past a closed plate raises RPM. The IAC steps closed to fight it. If the leak is larger than the IAC's authority, RPM stays high and an idle code can set. Fix the leak before any idle relearn. A relearn on a leaking engine memorizes the leak.

Lean mixture — possible, not automatic

Because the MAP sensor sees the leak, fuel is added with it. The mixture can stay near normal while the idle is wrong. A lean result is still POSSIBLE if the leak, a weak fuel supply, or an injector outruns that correction. Prove it with upstream O2 voltage and fuel trims at idle and at 2,500 rpm. TEST REQUIRED

What a leak can contribute to a glowing manifold

If the added air is not fully matched with fuel, combustion temperature rises and exhaust temperature rises with it. Iron that gets hot enough glows orange or red. That path is physically real. On this truck it is one POSSIBLE path. A misfire that burns fuel in the pipe, retarded timing, and a restricted catalyst can glow the same manifold. Guide 05. Do not close the case at "we found a split hose" if the manifold has been glowing. Confirm the mixture and look for a misfire before you drive it.

Do not spray a flammable leak detector on this truck yet

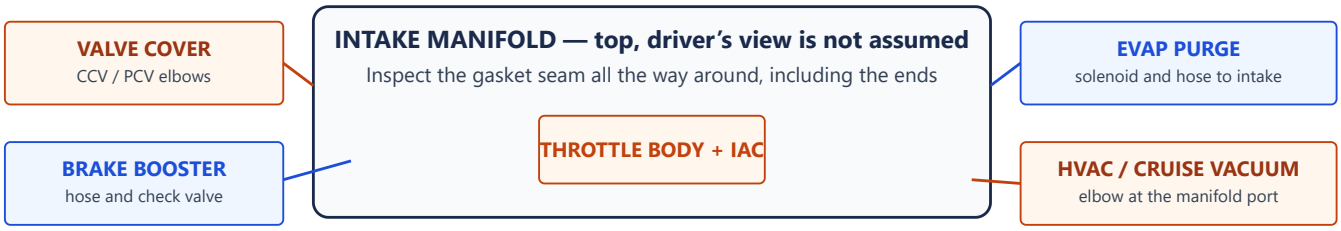
Carb cleaner, brake cleaner, and propane are fire hazards. This vehicle has an oil-on-exhaust concern and a glowing-manifold concern. Those sprays are not the test. Use your eyes, a vacuum gauge, scan data, and a smoke machine. A fire extinguisher stays within reach anyway.

EVAP note: a purge valve that sticks open can act as a vacuum leak and can pull fuel vapor or canister air into the intake. The idle may rise and the trims may swing rich or lean depending on what is in the canister. That is POSSIBLE. Test the valve. Do not replace the canister as a guess.

VACUUM AND EVAP · WHERE TO LOOK

Brittle hoses and elbows

These are common crack points on a 20-year-old 4.0L. **LIKELY** places to inspect. A place on this list is not a leak until you see the crack, hear the hiss, or watch smoke come out of it. Engine cold for the hands-on inspection.



Place	What age does to it	What you should see if it is the leak
CCV / PCV elbows at the valve cover and at the intake or air duct	Hard, split, or swollen from oil vapor. The short rear hose and the front elbow are repeat failures. **LIKELY**	A crack you can open with a finger, or smoke at the fitting. Dorman 47128 is only the short hose set. Compare it. Guide 09.
Hard plastic vacuum lines and molded elbows	They snap instead of bending. Tees crumble.	A white stress crack, or a line that is loose on a port.
Brake-booster hose and check valve	Soft spots, a split at the nipple, a check valve that rattles apart.	A hiss at the hose, a hard pedal, or RPM that falls when that hose is pinched with proper pliers. Pinch only if the hose will not crack.
Intake-manifold gasket	Common on high-mileage 4.0L engines, especially at the ends and lower corners. **LIKELY** pattern.	Smoke at the head-to-manifold seam, or a hiss you can move along the flange with a hose to your ear.
Throttle-body gasket and IAC O-ring	Left off, pinched, or flattened during an IAC job.	Smoke at that joint. Directly relevant if the IAC was recently handled.
EVAP purge hose and solenoid ports	Cracked ports, a hose pulled off, a valve stuck open.	Smoke out of the purge line with the valve closed, or idle that changes when the purge connector is unplugged. **TEST REQUIRED**
HVAC and cruise elbow at the intake	A 2004 factory action covers a narrow build window, 31 March–6 April 2004, where an under-cured elbow can collapse and block HVAC or cruise vacuum. Part in that letter: 55115901AG, elbow and check valve only. Outside that VIN window the same elbow still gets hard and can crack. **POSSIBLE**	A collapsed elbow starves HVAC and cruise. It does not raise idle. A cracked one can. Tell them apart.

VACUUM AND EVAP · SAFE TESTS

Safe leak tests, in order

1. **Cold visual and flex test.** With the engine off and cool, bend every elbow and short hose. Replace anything cracked, oil-soft, or so hard it has gone white at a fitting. Tug each connection. A hose that stays on only by habit will leak when the engine rocks.
2. **Listen, after the fire risk is down.** Only if the manifold is not glowing and oil has been cleaned off the exhaust. A length of hose used as a stethoscope along the intake seam, the booster hose, and the purge line. A sharp hiss that stops when you press a fitting is a find.
3. **Smoke test, engine off.** This is the preferred test. Introduce smoke into a manifold vacuum port. Block the intake inlet lightly so the smoke stays in the manifold. Watch the booster hose, the EVAP line, the intake gasket seam, the IAC base, the throttle-body gasket, and every CCV fitting. A smoke machine uses an inert or low-temperature vapor. It is the right tool while oil and heat are part of the story.
4. **Vacuum gauge, warm idle, only after a safe start is allowed.** Near this elevation a healthy 4.0L typically shows about **17–21 inHg**, steady. LIKELY screen, not a factory limit. A needle that flickers can be a leak or a misfire. A low steady reading can be a leak, late timing, low compression, or a restricted exhaust. Record the number. Do not condemn a gasket on the gauge alone.
5. **Scan data.** MAP, RPM, short-term and long-term fuel trims at idle and at about 2,500 rpm, upstream O2 voltage. On speed-density, a leak often shows up as high RPM with MAP higher than a normal idle and trims that are not dramatically positive. Trims that are largely positive only at idle, and come back at higher RPM, still suggest a small leak or an exhaust leak ahead of an upstream sensor. Write the numbers on the test record. TEST REQUIRED
6. **Pinch test, selective.** Pinch one suspect hose at a time with hose pliers while watching RPM. A drop in RPM means that hose was admitting air. Do not pinch a hard plastic line. Do not pinch the booster hose if the pedal is your only brake and the engine is the only source of assist you are relying on while the truck is not secured. Wheels chocked, Park, parking brake set.

Not approved on this job

Spraying carburetor cleaner or brake cleaner around the intake. Introducing propane under the hood. Those methods can find a leak on a clean, cool engine. This engine has an oil film and a possible extreme exhaust temperature. Skip them. If someone already did it, do not repeat it.

A found leak is still one finding

Repair it, then install the new IAC if the pintle was moved, clean the throttle body, and do the relearn. Verify idle. If the manifold glowed before the repair, watch it on the first idle from outside the wheel well, extinguisher ready, and shut down if color returns to the iron. Guide 05.

VACUUM AND EVAP · PCV, CCV, AND EVAP PARTS

PCV, CCV, and EVAP

The 4.0L cranksase ventilation is a hose-and-elbow system, not a single round PCV valve hidden under a rubber grommet the way older engines were. Oil vapor moves from the valve cover to the intake. The elbows harden, split, and then become both a vacuum leak and an oil leak onto the manifold. That ties Guide 04 to Guide 06.

CCV / PCV — inspect all of it

- Front elbow at the valve cover.
- Rear hose and grommet at the valve cover. This is the piece people find split.
- The hose from the cover to the intake or to the air duct.
- Any small hose that also tees toward the MAP sensor. Dorman 47128 is catalog-listed as that short dual-hose set, about 2.56 in, two hoses, OE cross 4854265. **CATALOG LISTED** It is not the whole system.
- Grommets that are loose in the cover. A new hose on a loose grommet still leaks.

EVAP — inspect, don't shotgun

- Purge solenoid on the engine. Cracked plastic ports are common once the hose has been yanked.
- Hose from the solenoid to the intake port. Hard, split, or off the nipple.
- Hose from the solenoid back toward the canister. A break there can set EVAP codes and may not change idle. Different fault.
- A stuck-open purge valve is an idle-air path. Unplug it only as a comparison, with the scan tool recording RPM. If idle falls and stays down, test or replace the valve. **TEST REQUIRED**
- Do not replace the charcoal canister, the leak-detection pump, or the fuel-tank seals because of P0505. Those parts belong to an EVAP-code diagnosis.

Repair rules

1. Replace the failed piece with hose rated for fuel and oil vapor, the right inside diameter, and a length that does not kink. A hardware-store vacuum hose of the wrong size is another leak.
2. Match the part in your hand to the catalog picture before you buy. Guide 09. Verify by VIN. The number 53030006 did not check out as a Mopar elbow for this Jeep. **NOT CONFIRMED**
3. Do not block a PCV port to "cure" a high idle. The crankcase then pressurizes, oil comes out of the next seal, and that oil lands on the exhaust.
4. After the hoses are sealed, smoke-test once more. Then, and only then, install the new IAC if it is not already on, and do the relearn in Guide 03.

MAP sensor. On this engine it is typically mounted in the intake, not on a long hose. Look for a cracked port or a sensor that is loose in the bore. A vacuum hose you "should" find may not exist on this manifold. Do not add one. If the sensor or its seal is leaking, smoke will show it. A MAP sensor that reads wrongly can also destabilize idle. Compare key-on MAP (it should read barometric, roughly 98–100 kPa near this elevation, before the engine starts) with idle MAP. **TEST REQUIRED** A failed MAP is **POSSIBLE** and is not the leading theory on a truck whose IAC pintle was pushed.